

Tangential Eigenvalue Universality on Constant-Curvature Surfaces

Sandbox for Paper I, Item #2

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Main result

Theorem 1 (Riemannian Havelock universality). *On the unit 2-sphere, the tangential eigenvalue of the constrained Hessian $\nabla^2(H_{\text{sph}} - \Omega J)$ at mode $m \in \{1, \dots, N-1\}$ is*

$$\hat{e}_t^T \nabla^2(H_{\text{sph}} - \Omega J) \hat{e}_t = \frac{m(N-m)}{2\xi},$$

identical to the flat-plane result at ring radius $\sqrt{\xi}$. The same holds on the hyperbolic plane with $(1+\xi)$ replaced by $(1-\xi)$ throughout.

Proof. Decompose $H_{\text{sph}} = H_{\text{flat}} + W$ where $H_{\text{flat}} = -\sum_{j < k} \ln |z_j - z_k|$ and $W = \frac{N-1}{2} \sum_k \ln(1+|z_k|^2)$. The Lagrangian is $\mathcal{L} = H_{\text{flat}} + W - \Omega J$ with $J = \sum_k (1-|z_k|^2)/(1+|z_k|^2)$. The tangential eigenvalue decomposes as

$$\hat{e}_t^T \nabla^2 \mathcal{L} \hat{e}_t = \underbrace{\hat{e}_t^T \nabla^2 H_{\text{flat}} \hat{e}_t}_{\text{flat Havelock}} + \underbrace{\hat{e}_t^T \nabla^2 (W - \Omega J) \hat{e}_t}_{\text{curvature + constraint}}.$$

We show the second term equals $(N-1)/(2\xi)$, i.e. the flat-plane Lagrange shift $-2\mu_L = (N-1)/(2\xi)$ at ring radius $\sqrt{\xi}$. Combined with the flat-plane unshifted tangential eigenvalue $(m(N-m) - (N-1))/(2\xi)$ (derived in the constrained-minimum proof, §6.2), this gives $m(N-m)/(2\xi)$.

Step 1: Tangential projection of $\nabla^2 W$. At vertex k on the ring ($|z_k|^2 = \xi$, angle θ_k), the Hessian of $W_k = \frac{N-1}{2} \ln(1+|z_k|^2)$ has entries

$$\begin{aligned} \frac{\partial^2 W_k}{\partial x_k^2} &= \frac{N-1}{1+\xi} - \frac{2(N-1)\xi \cos^2 \theta_k}{(1+\xi)^2}, & \frac{\partial^2 W_k}{\partial y_k^2} &= \frac{N-1}{1+\xi} - \frac{2(N-1)\xi \sin^2 \theta_k}{(1+\xi)^2}, \\ \frac{\partial^2 W_k}{\partial x_k \partial y_k} &= -\frac{2(N-1)\xi \sin \theta_k \cos \theta_k}{(1+\xi)^2}. \end{aligned}$$

The mode- m tangential unit vector at vertex k is $\hat{e}_{t,k} \propto \alpha_k (-\sin \theta_k, \cos \theta_k)$ with $\alpha_k = \cos(2\pi mk/N)$. The tangential projection at vertex k is

$$\begin{aligned} \sin^2 \theta_k \frac{\partial^2 W_k}{\partial x_k^2} - 2 \sin \theta_k \cos \theta_k \frac{\partial^2 W_k}{\partial x_k \partial y_k} + \cos^2 \theta_k \frac{\partial^2 W_k}{\partial y_k^2} \\ = \frac{N-1}{1+\xi} - \frac{2(N-1)\xi}{(1+\xi)^2} [\sin^2 \theta_k \cos^2 \theta_k - 2 \sin^2 \theta_k \cos^2 \theta_k + \cos^2 \theta_k \sin^2 \theta_k] \\ = \frac{N-1}{1+\xi}. \end{aligned}$$

The bracketed expression vanishes identically ($1 - 2 + 1 = 0$), so the projection is θ_k -independent and hence *mode-independent*. Summing over vertices with $\sum_k \alpha_k^2 = N/2 = \|\hat{e}_t\|^2$:

$$\hat{e}_t^T \nabla^2 W \hat{e}_t = \frac{N-1}{1+\xi}. \quad (1)$$

Step 2: Tangential projection of $\nabla^2 J$. With $J_k = (1 - |z_k|^2)/(1 + |z_k|^2)$:

$$\begin{aligned} \frac{\partial^2 J_k}{\partial x_k^2} &= -\frac{4}{(1+\xi)^2} + \frac{16\xi \cos^2 \theta_k}{(1+\xi)^3}, & \frac{\partial^2 J_k}{\partial y_k^2} &= -\frac{4}{(1+\xi)^2} + \frac{16\xi \sin^2 \theta_k}{(1+\xi)^3}, \\ \frac{\partial^2 J_k}{\partial x_k \partial y_k} &= \frac{16\xi \sin \theta_k \cos \theta_k}{(1+\xi)^3}. \end{aligned}$$

The same tangential projection yields

$$-\frac{4}{(1+\xi)^2} + \frac{16\xi}{(1+\xi)^3} [\sin^2 \theta \cos^2 \theta - 2 \sin^2 \theta \cos^2 \theta + \cos^2 \theta \sin^2 \theta] = -\frac{4}{(1+\xi)^2},$$

again θ -independent. Therefore

$$\hat{e}_t^T \nabla^2 J \hat{e}_t = -\frac{4}{(1+\xi)^2}. \quad (2)$$

Step 3: Combining $W - \Omega J$. With $\Omega = (N-1)(1-\xi^2)/(8\xi) = (N-1)(1-\xi)(1+\xi)/(8\xi)$:

$$\begin{aligned} \hat{e}_t^T \nabla^2 (W - \Omega J) \hat{e}_t &= \frac{N-1}{1+\xi} + \frac{(N-1)(1-\xi)(1+\xi)}{8\xi} \cdot \frac{4}{(1+\xi)^2} \\ &= \frac{N-1}{1+\xi} + \frac{(N-1)(1-\xi)}{2\xi(1+\xi)} \\ &= \frac{N-1}{1+\xi} \left[1 + \frac{1-\xi}{2\xi} \right] = \frac{N-1}{1+\xi} \cdot \frac{1+\xi}{2\xi} = \frac{N-1}{2\xi}. \end{aligned}$$

This equals exactly $-2\mu_L = (N-1)/(2\xi)$, the flat-plane Lagrange shift at ring radius $\sqrt{\xi}$.

Step 4: Assembly. The flat-plane unshifted tangential eigenvalue at radius $\sqrt{\xi}$ is $(m(N-m) - (N-1))/(2\xi)$ (from the constrained-minimum derivation). Adding the curvature-plus-constraint contribution $(N-1)/(2\xi)$:

$$\hat{e}_t^T \nabla^2 \mathcal{L} \hat{e}_t = \frac{m(N-m) - (N-1)}{2\xi} + \frac{N-1}{2\xi} = \frac{m(N-m)}{2\xi}.$$

Hyperbolic case. On $\mathbf{H}^2$ with $W = +(N-1)/2 \sum_k \ln(1 - |z_k|^2)$ and $J = \sum_k (1 + |z_k|^2)/(1 - |z_k|^2)$ (the Poincaré-disk formulation), the same tangential-projection argument gives $\hat{e}_t^T \nabla^2 W \hat{e}_t = -(N-1)/(1-\xi)$ and $\hat{e}_t^T \nabla^2 J \hat{e}_t = +4/(1-\xi)^2$, and with $\Omega_{\mathbf{H}^2} = -(N-1)(1+\xi)(1-\xi)/(8\xi)$ the combination gives

$$-\frac{N-1}{1-\xi} + \frac{(N-1)(1+\xi)(1-\xi)}{8\xi} \cdot \frac{4}{(1-\xi)^2} = -\frac{N-1}{1-\xi} + \frac{(N-1)(1+\xi)}{2\xi(1-\xi)} = \frac{N-1}{2\xi}.$$

□

1 Numerical verification

All intermediate identities are verified in `numerical_check.py` (this directory):

- $\hat{e}_t^T \nabla^2 W \hat{e}_t = (N - 1)/(1 + \xi)$: confirmed numerically for $N \in \{5, 6, 7\}$, $\xi \in \{0.2, 0.5\}$, all modes $m \in \{2, \dots, N - 2\}$ (mode-independent to 10^{-2}).
- $\hat{e}_t^T \nabla^2 J \hat{e}_t = -4/(1 + \xi)^2$: same parameters, mode-independent to 10^{-2} .
- $\hat{e}_t^T \nabla^2 (W - \Omega J) \hat{e}_t = (N - 1)/(2\xi)$: confirmed for $N \in \{5, 6, 7\}$, $\xi \in \{0.15, 0.3, 0.5, 0.7\}$.
- Full tangential eigenvalue $= m(N - m)/(2\xi)$: confirmed for $N \in \{4, 5, 6, 7\}$, $\xi \in \{0.15, 0.3, 0.5, 0.7\}$, all modes.
- Symbolic: the cross-term cancellation $\sin^2 \theta \cos^2 \theta - 2 \sin^2 \theta \cos^2 \theta + \cos^2 \theta \sin^2 \theta = 0$ is verified by `sympy` in the explicit tangential projection of $\nabla^2 W$.